

The Fungal BGC Atlas: a curated, evidence-linked knowledge base of 609 literature- and MIBiG-derived fungal biosynthetic gene cluster dossiers

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Abstract

Fungal biosynthetic gene clusters (BGCs) underlie a chemically and biologically important class of secondary metabolites, yet the primary evidence for any given cluster — which genes were actually shown to be required, which metabolites were directly observed, under what conditions, and how confidently — is scattered across decades of literature and only partially captured by curated sequence repositories such as MIBiG. We present the **Fungal BGC Atlas**, a curated, evidence-linked knowledge base of 609 fungal BGC dossiers, each anchored to a MIBiG accession and built by large-language-model-assisted extraction from primary literature followed by full manual review of every dossier. Beyond the cluster-level record, the Atlas resolves each dossier into 1572 genes, 807 metabolites, 915 experiments, 936 measurements, 1534 claims and 1264 curated subject–predicate–object relationships, all traceable to 2575 source records through 47 768 explicit evidence links. Structural and functional context is captured along 27 independent facet axes (biosynthesis class, chemistry, ecology, genome context and activation conditions), and 995 conflicts or unresolved uncertainties are preserved as first-class records rather than collapsed into a forced consensus. We package the Atlas as a relational SQLite database, a knowledge graph, and a self-contained interactive web dashboard (Overview, Taxonomy, Chemistry, Facet Explorer, Knowledge

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Graph, Evidence, Conflicts, a searchable Dossier Browser with cross-table detail views, and a live Data Dictionary), deployed on GitHub Pages and archived on Zenodo with an open licence (CC-BY-4.0). The Atlas turns 609 individually FAIR-adjacent literature records into a single queryable, comparable, and citable resource for fungal natural-product genome mining.

Database URL: <https://dr-richard-barker.github.io/fungal-bgc-atlas/>

Keywords: biosynthetic gene clusters; fungal natural products; MIBiG; knowledge graph; FAIR data; curated database; secondary metabolism; evidence provenance

1 Introduction

Fungi are prolific producers of secondary metabolites – polyketides, non-ribosomal peptides, terpenoids and hybrids thereof – with roles spanning virulence, chemical ecology, and pharmacology, from aflatoxins and other mycotoxins to clinically important natural products. The genes responsible for a given metabolite are typically physically clustered in the genome as a biosynthetic gene cluster (BGC), and the Minimum Information about a Biosynthetic Gene cluster (MIBiG) standard and its accompanying database have become the community reference point for recording which BGCs have been characterised and what they are reported to produce [1].

A MIBiG accession, however, is deliberately a minimal record: cluster boundaries, a product name, and a small set of controlled fields. The primary evidence behind that record – which individual gene was actually disrupted and shown to be required, which metabolite congener was directly detected versus inferred, under what culture conditions, with what caveats, and whether independent studies agree – lives only in the cited literature, in a different format for every paper, and is not itself queryable. This creates a substantial gap between what is nominally known about a fungal BGC and what can be programmatically retrieved, compared across BGCs, or checked for internal disagreement.

[TODO: Introduction – add 2–4 discipline-specific citations here once verified (e.g. antiSMASH/BGC genome-mining tools, prior fungal-BGC or natural-product knowledge-graph resources, and any of the 609 dossiers’ own source papers that motivate specific examples). None are included yet because fabricating bibliographic details from memory is deliberately avoided; see manuscript/latex/references.bib for a note on this.]

The FAIR principles – that data and the tools built on them should be Findable, Accessible, Interoperable and Reusable [2] – provide a natural design target for closing this gap: rather than re-reading primary literature for every comparative question, the evidence itself should be captured once, in a documented schema, with explicit provenance and with disagreement preserved rather than silently resolved.

Here we present the **Fungal BGC Atlas**, a resource that applies this design to 609 fungal BGC dossiers, each linked to a MIBiG accession. Every dossier was built by large-language-model (LLM)-assisted extraction from the primary literature and MIBiG record, followed by full manual review of all 609 dossiers by the curator (Methods). The Atlas resolves each dossier into its constituent genes, metabolites, experiments, measurements, claims and relationships, all linked back to their supporting sources, and explicitly records

the facets, uncertainties and unresolved normalisation questions that a single-value schema would otherwise discard. We describe the resource’s construction and content, summarise its composition, and present an interactive web dashboard and archived data/code package that make the Atlas directly usable for cross-BGC comparison and genome-mining hypothesis generation.

2 Materials and methods

2.1 Data curation

Each of the 609 dossiers originates from a MIBiG accession and its associated primary reference(s). Structured fields (BGC identity and boundaries, organism and strain, genes, metabolites, experiments, measurements, claims, relationships, and source citations) were extracted from the primary literature and the MIBiG record using Claude (Anthropic) large language models **[TODO: Methods – state the exact Claude model version(s) and extraction date range once confirmed; left as a placeholder rather than guessed]**. Every one of the 609 dossiers was then **fully manually reviewed** by the curator against the cited literature before being included in this snapshot; this is a completeness property of the current release, not a partial-sample audit.

Extraction outputs were passed through a normalisation layer (the `TERM` vocabulary and `NORMALIZATION_QUEUE` tables) that reconciles free-text labels for experimental conditions, interventions, assays and taxonomy against a controlled vocabulary where possible. Values that could not yet be confidently normalised (composite labels, type mismatches, or terms awaiting review) are retained rather than forced into a canonical bucket, and are tracked explicitly: 882 `TERM` rows remained in the normalisation queue at the time of this snapshot (Results). Similarly, where independent sources disagreed, or a boundary or causal claim was ambiguous, this was recorded as a `CONFLICT` row with an explicit issue type and status rather than resolved by editorial fiat – 995 such records exist in the current snapshot, of which 220 remain explicitly *open*.

2.2 Database construction

All code is released under `scripts/` in the project repository and is orchestrated by a single entry point, `run_all.py`, so the entire resource is deterministically rebuildable from the source workbook:

1. `parse_xlsx.py` reads the 23 data tables and 5 documentation sheets from the source workbook

and writes one clean CSV per table.

2. **build_database.py** loads the parsed tables into a documented SQLite database (`data/db/fungal_bgc_atl` schema in `db/schema.sql`), with every table's row count asserted against the source workbook's own summary counts as a build-time regression check.
3. **build_graph.py** constructs two graphs: a literal knowledge graph of the curated subject–predicate–object RELATIONSHIP triples, and an entity-linkage graph joining BGC, organism, gene, metabolite and source nodes on `dossier_id`, exported as GraphML, JSON and CSV edge/node lists.
4. **compute_stats.py** computes every descriptive statistic used in this manuscript and in the dashboard directly from the parsed tables, and emits both CSV and L^AT_EX-ready table snippets (`tables/`).
5. **make_figures.py** renders dated, publication-quality figure snapshots (300 dpi PNG + vector SVG) to `figures/<date>/`, with an accompanying manifest.
6. **build_dashboard.py** renders the interactive dashboard (`docs/index.html`) described in Results.

The relational schema comprises 23 tables sharing `dossier_id` as a common join key (Table 1 of the online Data Dictionary panel; full schema in `db/schema.sql`), spanning identity and administration (BGC, ORGANISM, STRAIN), independent categorisation (BGC_FACET, 27 axes), molecular content (GENE, METABOLITE), experimental context (EXPERIMENT, TERM, EXPERIMENT_LINK), results (MEASUREMENT, OBSERVATION), interpretive records (CLAIM, RELATIONSHIP, CONFLICT), provenance (SOURCE, EVIDENCE_LINK, ASSERTION), and curation-audit tables (COMPACT_FACT, COVERAGE, DOSSIER_SCHEMA, NORMALIZATION_QUEUE, PROVENANCE_EXCLUDED, RAW_ROW).

The interactive dashboard is a single self-contained HTML page using Plotly for charting and a small vanilla-JavaScript component for the searchable dossier browser, requiring no server or build step to view. A GitHub Actions workflow rebuilds the database, graph, statistics, figures and dashboard on every push to the main branch and deploys the result to GitHub Pages, so the published resource is always reproducible from the checked-in source data and code.

3 Results

3.1 Resource composition

The current snapshot (21 August 2026) integrates **609 fungal BGC dossiers**, each linked to a MIBiG accession, spanning primary references from 1991 to 2024 (Table 1, Fig. 3). The dossiers cover **371 curated organisms** and 1018 distinct strain labels; where taxonomy could be resolved, organisms span three phyla (167 Ascomycota, 10 Basidiomycota, 1 Mortierellomycota) and 68 genera, and of the 609 dossiers with a genus-resolved organism (283 of 609; genus is not yet resolved for the remaining organisms, Discussion), *Aspergillus* accounts for 113 dossiers, *Penicillium* 36, and *Fusarium* 17 (Fig. 1), consistent with the historical emphasis of fungal natural-product genome mining on these genera. Beyond the cluster level, the Atlas resolves 1572 genes, 807 metabolites, 915 experiments, 936 measurements and 1641 observations (Table 1).

Table 1: Fungal BGC Atlas: resource summary metrics.

Metric	Value	Notes
BGC dossiers	609	All linked to MIBiG accessions
Unique organisms	371	3 phyla, 68 genera
Strain labels	1018	
Genes	1572	
Metabolites	807	
Experiments	915	
Measurements	936	
Claims	1534	
Relationships (KG triples)	1264	
Conflicts / uncertainties	995	preserved, not collapsed
Sources	2575	
Evidence links	47768	record <-> source provenance
Publication span	1991-2024	

3.2 Biosynthetic and chemical diversity

MIBiG-reported BGC classes are dominated by polyketide synthase (PKS) and non-ribosomal peptide synthetase (NRPS) clusters: of the 20 most common raw class labels, “PKS (Unknown)” (63 dossiers), “NRPS (Type I)” (57) and “PKS (Type I)” (38) are the largest single categories, with a further 19 dossiers combining NRPS and PKS machinery and 18 assigned to terpene biosynthesis (Table 3, Fig. 2). Beyond this single class label, each BGC is independently characterised along 27 facet axes spanning biosynthesis, chemistry, ecology, genome context and activation conditions (BGC_FACET, 7800 rows), allowing a cluster to be queried

simultaneously by, for example, chemical class, host organism ecology, and the developmental state under which it is known to be expressed – without collapsing these into a single forced category.

3.3 Knowledge graph of curated relationships

The RELATIONSHIP table records 1264 curated statements of the form subject–predicate–object (e.g. gene *encoded_by* protein, cluster *produces* metabolite, intervention *required_for* phenotype); 994 of these have a complete triple and are rendered as the Atlas knowledge graph (1870 entity nodes, 994 typed edges), which can be filtered interactively by predicate in the dashboard (Fig. 4 of the online dashboard; Materials and methods). The most common predicates are *produces* (60 instances), *required_for* (31), *synthesizes* (28), *cyclizes* (27) and *encoded_by* (26), reflecting the underlying emphasis on causal, pathway-level evidence rather than purely descriptive annotation.

3.4 Evidence, provenance and preserved uncertainty

Every claim, measurement, gene function and relationship in the Atlas is traceable to a SOURCE record (2575 sources) through an explicit EVIDENCE_LINK (47 768 links), so a user can move from any field back to the literature or database record that supports it (Fig. 4). Rather than resolving disagreement editorially, 995 CONFLICT records preserve boundary disputes, strain- or assay-dependent discordance, and unresolved causal questions as first-class, queryable data; the most common issue types are ecological-role ambiguity (20 records), BGC boundary-scope disagreement (18) and depth-of-evidence questions about gene function (13) (Table 4, Fig. 5). Of the 995 conflicts, 220 remain explicitly open and 199 are marked resolved-for-this-Atlas, with the remainder spanning provenance- or curation-specific resolutions – the full distribution is available in `supplementary/supp_table9_conflict_status.csv`.

3.5 Interactive dashboard

The Atlas is published as a single self-contained web page (<https://dr-richard-barker.github.io/fungal-bgc-atlas/>) with nine linked panels: an **Overview** of headline metrics and per-dossier evidence-coverage completeness (Fig. 6); a **Taxonomy Explorer** (phylum–class–order–genus composition); a **Chemistry & Biosynthesis** panel switching between BGC class, chemical class, product family and biological role; a **Facet Explorer** over all 27 facet axes; the **Knowledge Graph** described above; an **Evidence**

& Provenance panel (source types, publication timeline, evidence-link density); a **Conflicts & Uncertainty** panel; a searchable **Dossier Browser** that assembles, for any of the 609 BGCs, its full cross-table record (organism, strain, genes, metabolites, experiments, measurements, claims, relationships, conflicts and sources, joined on `dossier_id`); and a live **Data Dictionary** rendered directly from the source workbook’s own schema documentation.

4 Discussion

The Fungal BGC Atlas is designed to complement, not replace, MIBiG: MIBiG remains the authoritative, minimal, sequence-anchored record of a BGC, while the Atlas captures the surrounding evidence – which gene was shown to matter, which metabolite was actually detected, under what conditions, by whom, and with what caveats – as structured, queryable, and independently citable data. By resolving each dossier into genes, metabolites, experiments, measurements, claims and typed relationships, all linked back to source records, cross-cluster questions that previously required re-reading the primary literature (which predicates most often co-occur with a given chemical class; which BGCs have unresolved boundary disputes; which experimental conditions are associated with a given activation state) become directly queryable against the SQLite database or the knowledge graph.

Three design choices follow the FAIR principles [2] directly. First, **Interoperability** is served by a documented relational schema in which every table shares `dossier_id` as a join key, so the entity-linkage and knowledge-graph views in the dashboard are direct database queries rather than bespoke integration code. Second, **Reusability** is protected by preserving disagreement and unresolved normalisation explicitly (995 conflict records, 882 open normalisation-queue rows) rather than silently forcing a single value, so a downstream user can see exactly where the curated record is contested or provisional. Third, **Findability and Accessibility** are served by packaging the database, knowledge graph, figures, tables, supplementary material and code together for Zenodo archival under an open licence, with a live interactive dashboard as the primary point of entry.

4.1 Limitations

Two limitations follow directly from the current snapshot’s construction and are stated explicitly rather than hidden. First, taxonomic resolution is incomplete: phylum could be resolved for 178 of 371 organisms

and genus for organisms underlying 283 of 609 dossiers at this snapshot; the remainder await taxonomy normalisation and are tracked in `NORMALIZATION_QUEUE` rather than assigned a placeholder value. Second, while every dossier was fully manually reviewed by the curator, this release reflects a single-curator review pass; broadening review to multiple independent curators, and formalising inter-curator agreement on the `CONFLICT` and normalisation-status fields, is a natural next step and is discussed as future work below. **[TODO: Discussion – add the exact LLM model version(s)/dates here alongside the limitation statement once confirmed (Methods).]**

4.2 Future work

The build pipeline (Materials and methods) is designed so that on-boarding additional BGCs is an append-and-rebuild operation rather than a schema change: new dossiers extend the same 23-table schema, are validated by the same row-count and referential-integrity checks, and immediately appear in the dashboard and knowledge graph. Natural extensions include: closing the normalisation queue (882 open `TERM` rows) through continued curator review; resolving the remaining open conflicts (220 of 995) as new evidence becomes available; cross-linking dossiers to external resources beyond MIBiG (e.g. chemical structure databases for the 807 curated metabolites); and opening the curation workflow to community contribution with the same evidence-linked, conflict-preserving schema enforced at ingestion.

5 Data availability

The Fungal BGC Atlas database, knowledge graph, source code, figures, tables and supplementary material are freely available under the Creative Commons Attribution 4.0 International licence (CC-BY-4.0) at:

- **Interactive dashboard:** <https://dr-richard-barker.github.io/fungal-bgc-atlas/>
- **Source repository (code + data + manuscript):** <https://github.com/dr-richard-barker/fungal-bgc-atlas>
- **Archived release (Zenodo):** **[TODO: Zenodo DOI – to be assigned when the repository is deposited; this manuscript is prepared alongside the Zenodo packaging (`.zenodo.json`, `CITATION.cff`) but the deposit itself is a separate step.]**

205 All code used to parse the source workbook, build the relational database and knowledge graph, com-
206 pute the statistics reported here, render the figures, and build the dashboard is provided in `scripts/` and
207 orchestrated by `run_all.py`, so the entire resource is deterministically reproducible from the checked-in
208 source data.

209 Availability of source code and requirements

- 210 • Project name: Fungal BGC Atlas
- 211 • Project home page: <https://github.com/dr-richard-barker/fungal-bgc-atlas>
- 212 • Operating system(s): Platform independent
- 213 • Programming language: Python 3.9+ (build pipeline); HTML/JavaScript (dashboard)
- 214 • License: CC-BY-4.0

215 Conflict of interest

216 None declared.

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222 **sources).]**

223 References

224 [1] Barbara R. Terlouw, Kai Blin, Jorge C. Navarro-Muñoz, Nicole E. Avalon, Marc G. Chevrette, Susan
225 Egbert, Sanghoon Lee, David Meijer, Marcus J. J. Recchia, Zachary L. Reitz, Jeffrey A. van Santen,

226 Nelly Selem-Mojica, Ewa Torés, Mohammad Alanjary, Sarah L. Robinson, Gregory Lund, Sam C.
227 Epstein, Auristela C. Sisto, Louise K. Charkoudian, Jerome Collemare, Roger G. Linington, Tilmann
228 Weber, and Marnix H. Medema. MIBiG 3.0: a community-driven effort to annotate experimentally
229 validated biosynthetic gene clusters. *Nucleic Acids Research*, 51(D1):D603–D610, 2023. doi: 10.1093/
230 nar/gkac1049.

231 [2] Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton,
232 Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E. Bourne, et al.
233 The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3:160018,
234 2016. doi: 10.1038/sdata.2016.18.